Assignment - 8

① Show that the inverse metric is a tensor of rank (2,0).

$$\boxed{\eta \eta^{-1} = \mathbb{I}}$$

Ans: If $\eta \longrightarrow \boxed{\eta'_{\mu\nu} = (\Lambda^{-1})^{\alpha}{}_{\mu} (\Lambda^{-1})^{\beta}{}_{\nu} \eta_{\alpha\beta}}$ — ①

Now, $\eta \eta^{-1} = \mathbb{I}$

$$\Rightarrow \eta^{\mu}{}_{\rho} (\eta^{-1})^{\rho}{}_{\nu} = \delta^{\mu}_{\nu}$$ (in matrix notation)

Let $\eta \longrightarrow \eta'$

$$\therefore \eta'^{\mu}{}_{\rho} (\eta'^{-1})^{\rho}{}_{\nu} \cancel{= \eta^{\mu}{}_{\rho} (\eta^{-1})^{\rho}{}_{\nu}} = \delta^{\mu}_{\nu}$$

$$\Rightarrow \left[(\Lambda^{-1})^{\alpha}{}_{\mu} (\Lambda^{-1})^{\beta}{}_{\rho} \eta_{\alpha\beta} \right] (\eta'^{-1})^{\rho}{}_{\nu} = \delta^{\mu}_{\nu}$$

$$\cancel{= \eta^{\mu}{}_{\rho} (\eta^{-1})^{\rho}{}_{\nu}}$$ [from ①]

$$\Rightarrow (\eta'^{-1})^{\rho}{}_{\nu} = \Lambda^{\alpha}{}_{\mu} \Lambda^{\beta}{}_{\rho} (\eta^{-1})_{\alpha\beta} \delta^{\mu}_{\nu}$$

$$\Rightarrow \boxed{(\eta'^{-1})_{\rho\nu} = \Lambda^{\alpha}{}_{\nu} \Lambda^{\beta}{}_{\rho} (\eta^{-1})_{\alpha\beta}}$$

As can be seen, there are 2 Λ indices in the transformat^n. Hence, it is a tensor of rank (2,0).

(2) Show that the charge conservation eqn is Lorentz invariant, i.e., if $J_\mu \xrightarrow{\Lambda} J'_\mu$, then

$$\partial'_\mu J'^\mu = 0$$

Ans: If $\partial_\mu \to \partial'_\mu = (\Lambda^{-1})^\alpha_{\ \mu}\, \partial_\alpha$ (co-vector) — (i)

If $J^\mu \to J'^\mu = \Lambda^\mu_{\ \beta} J^\beta$ (contra-vector) — (ii)

$$\therefore \partial'_\mu J'^\mu = [(\Lambda^{-1})^\alpha_{\ \mu}\, \partial_\alpha]\,(\Lambda^\mu_{\ \beta} J^\beta) \quad [\text{from (i) \& (ii)}]$$

$$= (\Lambda^{-1})^\alpha_{\ \mu}\, \Lambda^\mu_{\ \beta}\, \partial_\alpha J^\beta$$

$$= \delta^\alpha_{\ \beta}\, \partial_\alpha J^\beta$$

$$= \partial_\alpha J^\alpha = 0$$

$$\Rightarrow \boxed{\partial'_\mu J'^\mu = 0} \quad \text{(Proved)}$$

(3) An observer in O-frame (at rest) finds $\bar{E} = (E_x, E_y, E_z)$ & $\bar{B} = (0, 0, 0)$. Find $\bar{E}'$ & $\bar{B}'$ in a moving frame moving with velocity v in the x-direction.

Ans: Maxwell's Tensor is

$$F_{\mu\nu} = \begin{pmatrix} 0 & -\frac{E_x}{c} & -\frac{E_y}{c} & -\frac{E_z}{c} \\ \frac{E_x}{c} & 0 & B_z & -B_y \\ \frac{E_y}{c} & -B_z & 0 & B_x \\ \frac{E_z}{c} & B_y & -B_x & 0 \end{pmatrix}.$$

$$\Lambda^{-1} = \begin{pmatrix} \gamma & +\frac{\gamma v}{c} & 0 & 0 \\ +\frac{\gamma v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\boxed{F'_{\mu\nu} = (\Lambda^{-1})^\alpha_{\ \mu}\, (\Lambda^{-1})^\beta_{\ \nu}\, F_{\alpha\beta}}$$

$\therefore \quad B_x' = F'_{23}$

$= (\Lambda^{-1})^{\alpha}{}_{2} (\Lambda^{-1})^{\beta}{}_{3} F_{\alpha\beta}$

$= F_{23}$ $\quad$ [$(\Lambda^{-1})^{2}{}_{2}$ & $(\Lambda^{-1})^{3}{}_{3}$ are the only non-zero components]

$= B_x = 0$

$$\boxed{B_x' = 0} \quad \text{— (i)}$$

$\therefore \quad B_y' = F'_{31}$

$= (\Lambda^{-1})^{\alpha}{}_{3} (\Lambda^{-1})^{\beta}{}_{1} F_{\alpha\beta}$

$= (\Lambda^{-1})^{\beta}{}_{1} F_{3\beta}$

$= (\Lambda^{-1})^{0}{}_{1} F_{30} + (\Lambda^{-1})^{1}{}_{1} F_{31}$

$= \left(\frac{\gamma v}{c}\right)\left(\frac{E_z}{c}\right) + \gamma B_y$ $\quad$ [$B_y = 0$]

$$\boxed{B_y' = \frac{\gamma v}{c^2} E_z} \quad \text{— (ii)}$$

$\therefore \quad B_z' = F'_{12}$

$= (\Lambda^{-1})^{\alpha}{}_{1} (\Lambda^{-1})^{\beta}{}_{2} F_{\alpha\beta}$

$\cancel{= (\Lambda^{-1})^{0}{}_{1} (\Lambda^{-1})^{\beta}{}_{2} F_{0\beta} + (\Lambda^{-1})^{1}{}_{1} (\Lambda^{-1})^{\beta}{}_{2} F_{1\beta}}$

$= \cancel{\left(\frac{\gamma v}{c}\right)[}\ (\Lambda^{-1})^{\alpha}{}_{1} F_{\alpha 2}$

$= \left(\frac{\gamma v}{c}\right)\left(-\frac{E_y}{c}\right) + \gamma B_z$

$$\boxed{B_z' = -\frac{\gamma v}{c^2} E_y} \quad \text{— (iii)}$$

$\therefore \quad \frac{E_x'}{c} = F'_{10}$

$= (\Lambda^{-1})^{\alpha}{}_{1} (\Lambda^{-1})^{\beta}{}_{0} F_{\alpha\beta}$

$= (\Lambda^{-1})^{0}{}_{1} (\Lambda^{-1})^{\beta}{}_{0} F_{0\beta} + (\Lambda^{-1})^{1}{}_{1} (\Lambda^{-1})^{\beta}{}_{0} F_{1\beta}$

$= \left(\frac{\gamma v}{c}\right)\left[\left(\frac{\gamma v}{c}\right)\left(-\frac{E_x}{c}\right)\right] + \gamma\left[\gamma \frac{E_x}{c}\right]$

$= \gamma^2 \frac{E_x}{c}\left[\gamma 1 - \frac{v^2}{c^2}\right] = \frac{E_x}{c}$

$$\Rightarrow \boxed{E_x' = E_x} \quad \text{— (IV)}$$

$$\therefore \frac{E_y'}{c} = F'_{20}$$

$$= (\Lambda^{-1})^{\alpha}{}_{2} \, (\Lambda^{-1})^{\beta}{}_{0} \, F_{\alpha\beta}$$

$$= (\Lambda^{-1})^{\beta}{}_{0} \, F_{2\beta}$$

[$(\Lambda^{-1})^{2}{}_{2}$ is the only non-zero component]

$$= \gamma \frac{E_y}{c} + \left(\frac{\gamma v}{c}\right)(-B_z)$$

$$\Rightarrow \boxed{E_y' = \gamma E_y} \quad \text{— (V)}$$

[$B_z = 0$]

$$\therefore \frac{E_z'}{c} = F'_{30}$$

$$= (\Lambda^{-1})^{\alpha}{}_{3} \, (\Lambda^{-1})^{\beta}{}_{0} \, F_{\alpha\beta}$$

[$(\Lambda^{-1})^{3}{}_{3}$ is the only non-zero component]

$$= (\Lambda^{-1})^{\beta}{}_{0} \, F_{3\beta}$$

$$= \gamma \frac{E_z}{c} + \left(\frac{\gamma v}{c}\right) B_y$$

$$\Rightarrow \boxed{E_z' = \gamma E_z} \quad \text{— (VI)}$$

[$B_y = 0$]

From (I) – (VI):

$$\bar{E}' = (E_x, \gamma E_y, \gamma E_z)$$

[Ans]

$$\bar{B}' = \left(0, \frac{\gamma v}{c^2} E_z, -\frac{\gamma v}{c^2} E_y\right)$$